

Supply chain inventory optimization

Less inventory, with a measurable service target

6 distribution centers / 15 SKUs / 104 weeks

26.6%

less modeled on-hand inventory

97.9%

modeled unit fill on the final 26 weeks

Set 90 stock targets around one service goal

Protect service

Fill at least 95% of requested units.

Release excess stock

Lower average physical inventory where the buffer adds little protection.

9,360

weekly records

90

DC-SKU pairs

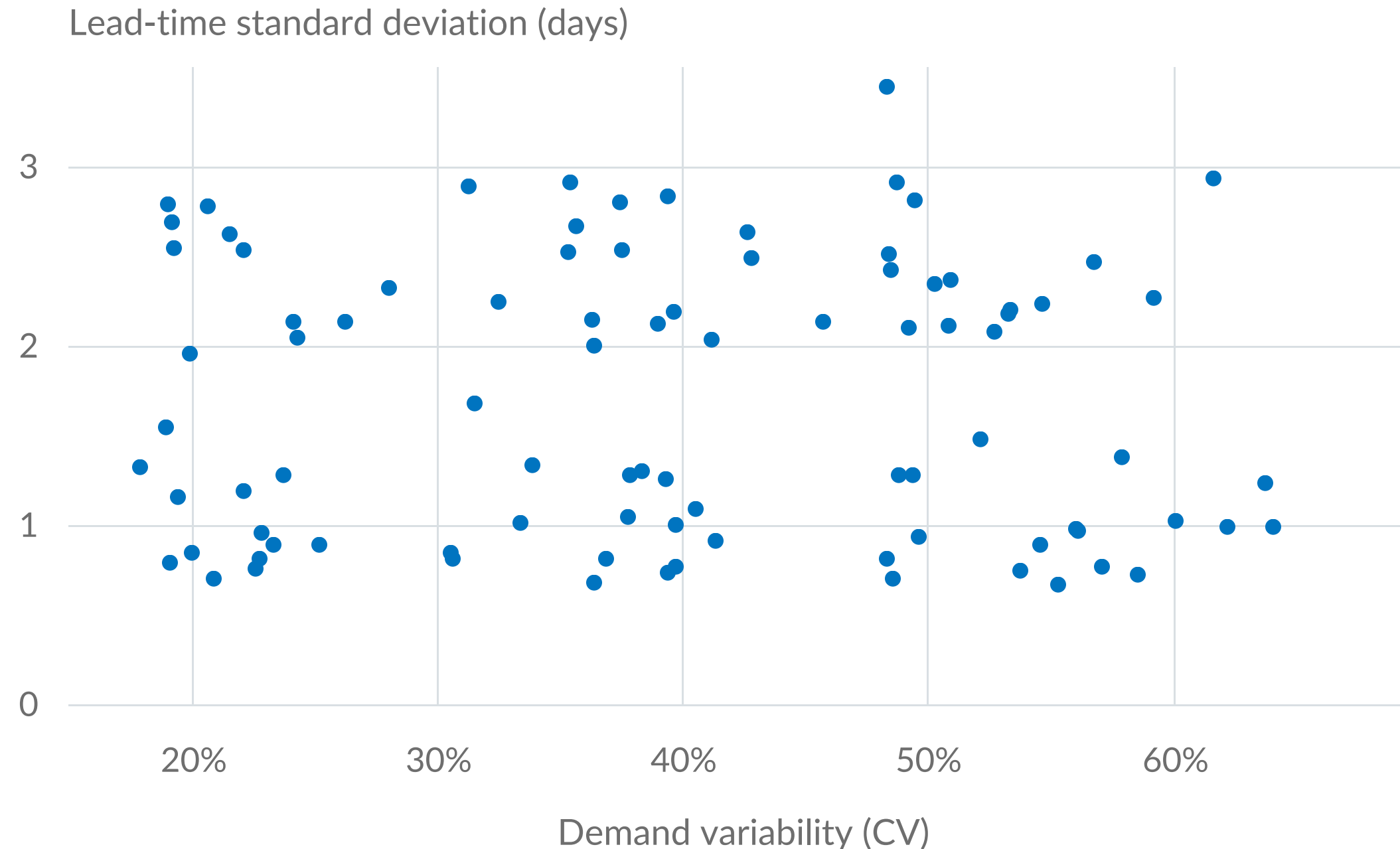
104

consecutive weeks

95%

unit fill target

Demand and lead-time uncertainty vary sharply by pair



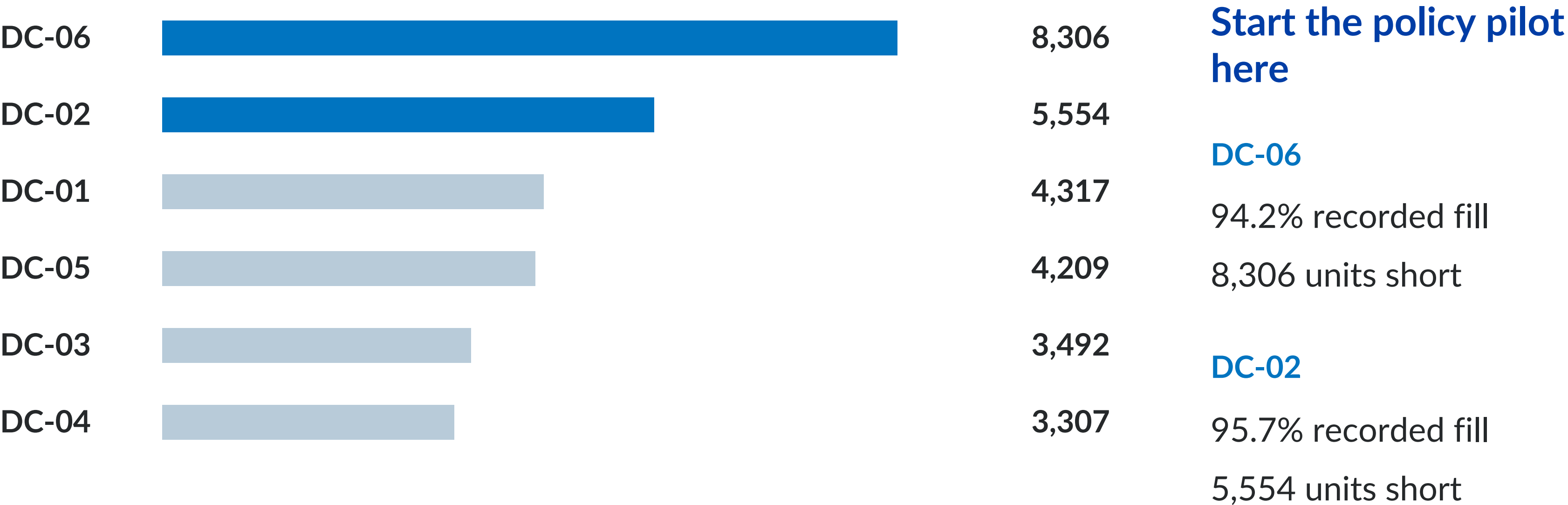
18%–64%

range of demand CV

Each pair has a constant safety-stock setting across all 104 weeks.

DC-06 and DC-02 account for 47.5% of unmet demand

Unmet units in the final 26 weeks



Cover demand through the next replenishment window

01

Learn each pair

52 observed weeks of demand
and lead time

02

Sample protection demand

10,000 simulated totals over lead
time + review

03

Set the target

95th percentile of protection
demand

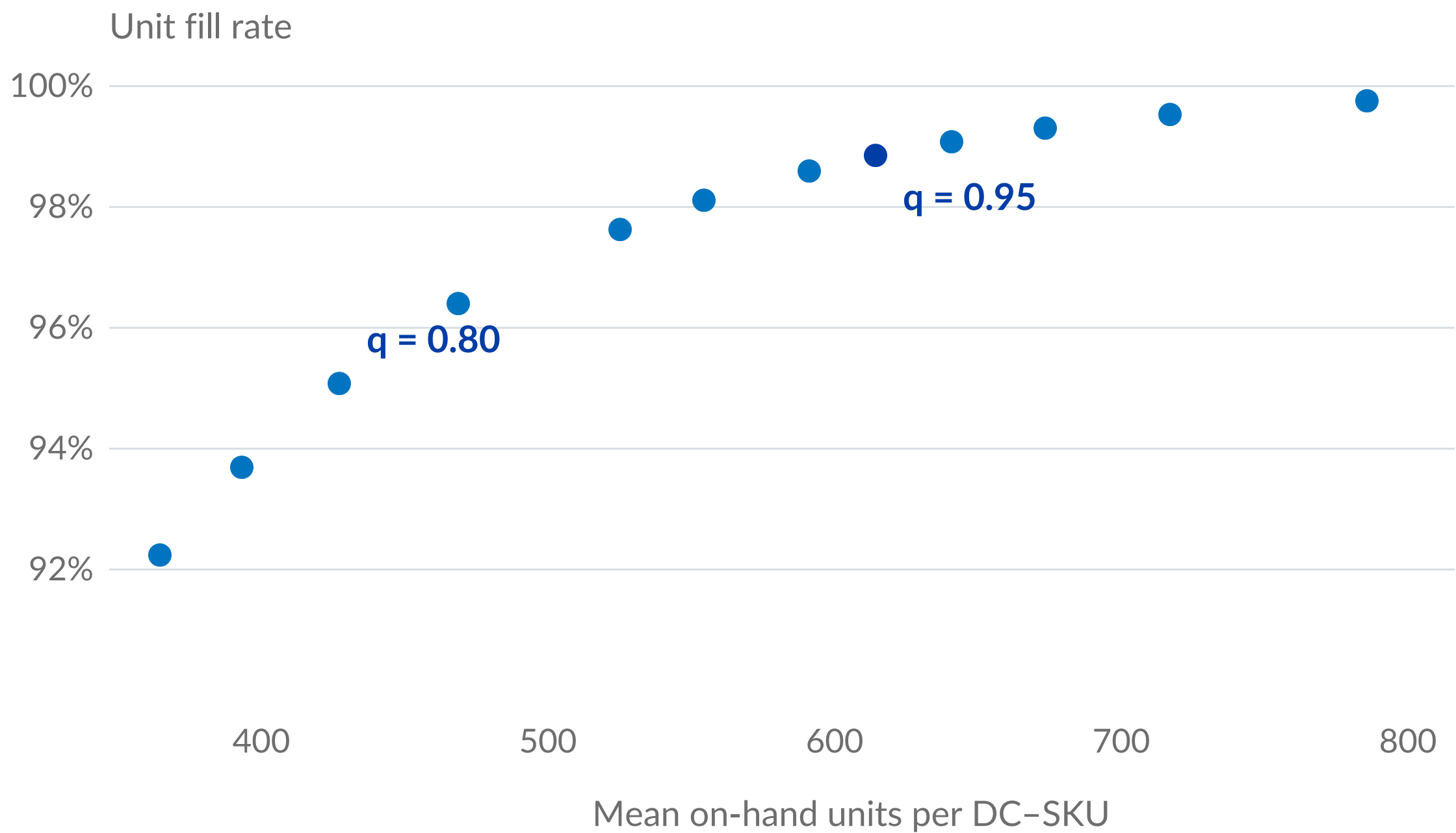
Weekly order = $\max(0, \text{target} - \text{on-hand} - \text{open orders})$

Safety stock = target – expected protection demand, floored at zero.

Time order is preserved; the final estimate is exploratory



More protection improves service, but costs inventory



95th-percentile policy

98.8%

validation unit fill

617

average on-hand units per pair

26.6% less stock, with 97.9% of units filled

Average on-hand units per DC-SKU

Fixed safety stock



748

Recommended



549

UNIT FILL

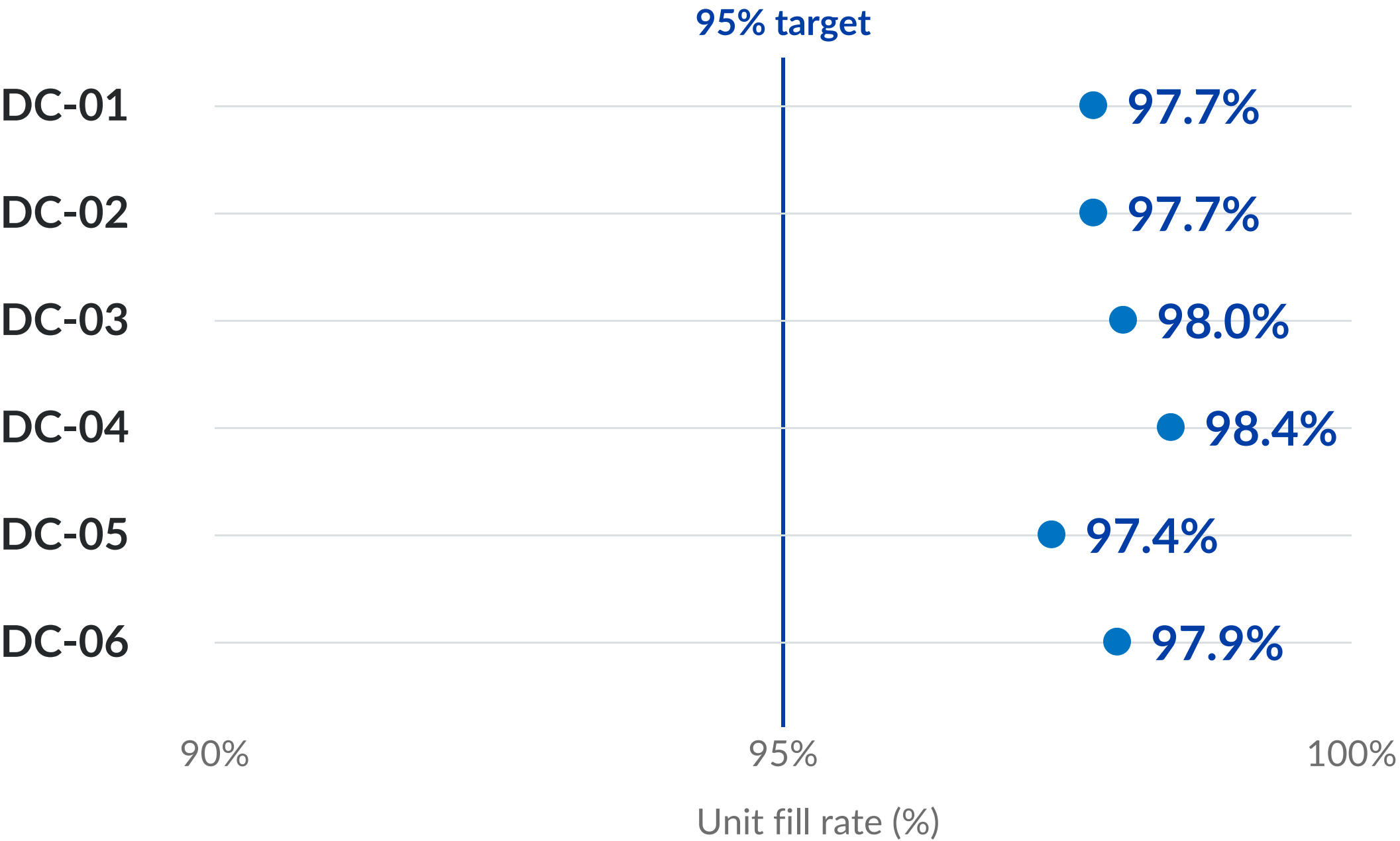
99.4% → 97.9%

STOCKOUT WEEKS

1.5% → 6.5%

The inventory reduction carries a service cost: 1.6 percentage points of unit fill.

Every DC clears 95%; nine individual pairs still miss



9 / 90

pairs below 95% unit fill

91.7%

lowest pair fill rate

Buffers fall for 88 pairs, with two targeted increases

DC	SKU	Current buffer	New buffer	Order-up-to point
DC-06	SKU-SKU-402	964	359	1,675
DC-01	SKU-SKU-402	970	449	1,666
DC-06	SKU-502	958	438	1,727
DC-01	SKU-SKU-201	731	781	1,821
DC-06	SKU-SKU-102	368	381	903
DC-01	SKU-SKU-102	307	300	676

The order-up-to point covers expected demand plus the recommended buffer.

The reconstructed baseline outperforms recorded operations

96.2%

Actual recorded unit fill

Observed demand and shortages

99.4%

Modeled fixed-policy unit fill

Common assumed weekly order logic

The missing ordering rule is material.

Treat the stock reduction as a policy scenario. Verify receipts, timing and order constraints in a pilot.

Pilot the rule, then expand only when local service holds

01

Verify the inputs

Confirm inventory position, arrival timing and unit costs.

02

Pilot DC-06 + DC-02

Review orders every Monday; track service by DC-SKU.

03

Reassess monthly

Refresh demand and lead time; review weak pairs and exceptions.

Expansion gate: $\geq 95\%$ unit fill, acceptable inventory and verified operational feasibility.

Use consistent units and distinguish the two service measures

Unit fill rate

Total served units $\div$ total requested units.

Demand weighted across pairs.

Stockout-week rate

Pair-weeks with unmet demand $\div$ all pair-weeks.

A frequency measure, not a unit fill measure.

Mean on-hand

Average of start-after-receipts and end stock.

A weekly inventory proxy in units per pair.

Inventory position

On-hand stock plus scheduled open orders.

Lost sales; no backorder balance.

Protection window

One review week + $\text{ceil}(\text{lead time in days} \div 7)$.

Orders arrive at the start of the arrival week.